

Stashio E-commerce Platform

Software Engineering Project Documentation

1. Problem Statement & Objectives

1.1 Problem Statement

Traditional retail and small-to-medium businesses face significant challenges in reaching a broader customer base due to high operational costs, limited physical presence, and outdated manual processes. Customers, on the other hand, lack a convenient, centralized platform to browse, compare, and purchase products from multiple vendors without visiting physical stores.

Stashio addresses these challenges by providing a fully digital, scalable e-commerce platform that connects buyers and sellers in a seamless online environment. The platform eliminates geographical barriers, reduces overhead costs, and empowers both vendors and customers through a modern, intuitive interface.

1.2 Project Objectives

The primary objectives of the Stashio platform are:

- Provide a centralized marketplace where customers can search, browse, and purchase products from multiple categories.
- Enable vendors/admins to manage product listings, inventory, and orders efficiently.
- Implement a secure, reliable payment processing system supporting multiple payment methods.
- Deliver a responsive user interface accessible on both desktop and mobile devices.
- Automate order tracking and notification to improve post-purchase customer experience.
- Ensure platform scalability to support growing numbers of users, products, and transactions.
- Maintain high data security standards to protect customer and transaction data.

1.3 Scope

The Stashio platform covers the following functional areas:

- User Management: Registration, authentication, profile management.
- Product Management: Listings, categories, search, filtering, and reviews.
- Shopping Cart & Checkout: Cart management, order placement, payment integration.

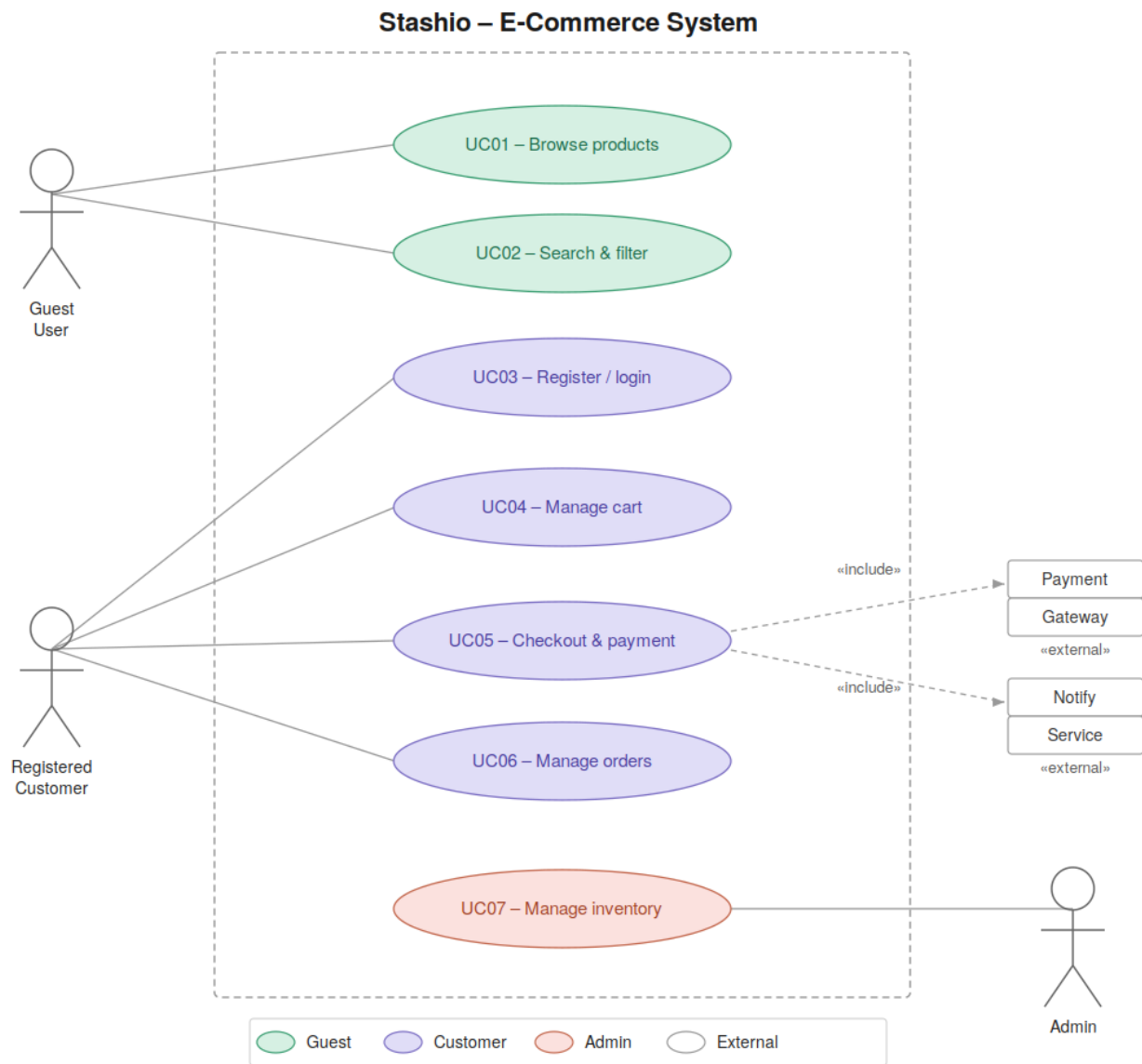
2. Use Case Diagram & Descriptions

2.1 System Actors

The Stashio system has the following primary actors:

- Guest User – An unregistered visitor who can browse products and view listings.
- Registered Customer – An authenticated user who can purchase products, manage orders, and write reviews.
- Administrator – A privileged user who manages the platform, products, users, and reports.
- Payment Gateway – External system that processes payment transactions.

2.2 Use Case Diagram



2.3 Use Case Descriptions

UC01 – Browse Products

Use Case ID	UC01
Use Case Name	Browse Products
Actor(s)	Guest User, Registered Customer
Description	The user navigates through product categories and views product listings on the platform.
Pre-conditions	Platform is accessible; product catalog is populated.
Main Flow	1. User opens the homepage. 2. User selects a product category. 3. System displays products in that category with thumbnail, name, and price. 4. User scrolls through the listing. 5. User clicks a product to view its detail page.
Post-conditions	User views product listing successfully.

UC02 – Search & Filter Products

Use Case ID	UC02
Use Case Name	Search & Filter Products
Actor(s)	Guest User, Registered Customer
Description	The user searches for products using keywords and applies filters to narrow results.
Pre-conditions	Platform is accessible.
Main Flow	1. User enters a keyword in the search bar. 2. System returns matching products. 3. User applies filters (price range, category, rating). 4. System refreshes results based on filters. 5. User views filtered results.
Post-conditions	Filtered search results are displayed accurately.

UC03 – Register / Login

Use Case ID	UC03
Use Case Name	Register / Login
Actor(s)	Guest User
Description	A guest creates a new account or logs in to an existing account.
Pre-conditions	User has not authenticated yet.

Main Flow	1. User clicks 'Sign Up' or 'Login'. 2. For Sign Up: user fills form (name, email, password) → system validates → account created. 3. For Login: user enters email & password → system authenticates → session started. 4. System redirects user to homepage as Registered Customer.
Post-conditions	User is authenticated and session is active.

UC04 – Manage Shopping Cart

Use Case ID	UC04
Use Case Name	Manage Shopping Cart
Actor(s)	Registered Customer
Description	Customer adds, updates, or removes products from the shopping cart.
Pre-conditions	Customer is logged in.
Main Flow	1. Customer views product detail page. 2. Customer clicks 'Add to Cart'. 3. System adds item to cart and updates cart count. 4. Customer opens cart to review items. 5. Customer updates quantity or removes items. 6. System recalculates total price.
Post-conditions	Cart reflects the latest items and quantities.

UC05 – Checkout & Payment

Use Case ID	UC05
Use Case Name	Checkout & Payment
Actor(s)	Registered Customer, Payment Gateway
Description	Customer proceeds to checkout, enters shipping details, and completes payment.
Pre-conditions	Customer has items in cart and is logged in.
Main Flow	1. Customer clicks 'Checkout'. 2. System displays order summary. 3. Customer enters/confirmes shipping address. 4. Customer selects payment method. 5. System sends payment request to Payment Gateway. 6. Payment Gateway processes and returns success/failure. 7. System creates order record and sends confirmation notification.
Post-conditions	Order is placed, payment confirmed, confirmation email sent.

UC06 – Manage Orders

Use Case ID	UC06
Use Case Name	Manage Orders

Actor(s)	Registered Customer
Description	Customer views order history, tracks current orders, and requests cancellations or returns.
Pre-conditions	Customer is logged in with at least one order.
Main Flow	1. Customer navigates to 'My Orders'. 2. System displays list of orders with status. 3. Customer selects an order to view details. 4. Customer may request cancellation (if status is Pending) or return (if Delivered). 5. System updates order status accordingly.
Post-conditions	Order status is updated and customer is notified.

UC08 – Manage Inventory (Admin)

Use Case ID	UC07
Use Case Name	Manage Inventory
Actor(s)	Administrator
Description	Admin adds new products, updates existing listings, and monitors stock levels.
Pre-conditions	Admin is authenticated.
Main Flow	1. Admin navigates to Dashboard → Products. 2. Admin adds a new product (name, description, images, price, stock, category). 3. System validates and saves the product. 4. Admin can edit or deactivate existing products. 5. System sends low-stock alerts when quantity drops below threshold.
Post-conditions	Product catalog is updated and reflects changes immediately.

3. Functional & Non-Functional Requirements

3.1 Functional Requirements

The following table lists the functional requirements of the ShopZone platform:

ID	Requirement	Description	Priority
FR01	User Registration	The system shall allow new users to register with name, email, and password.	High
FR02	User Authentication	The system shall authenticate users via email/password with session management.	High
FR04	Product Browsing	The system shall display products organized by categories with images, names, and prices.	High
FR05	Product Search	The system shall allow users to search products by keyword across name and description.	High
FR06	Product Filtering	The system shall allow filtering by category, price range, brand, and rating.	High
FR07	Product Details	The system shall display full product details including images, specs, reviews, and availability.	High
FR08	Shopping Cart	The system shall allow users to add, update quantity, and remove products from cart.	High
FR09	Checkout Process	The system shall guide users through a multi-step checkout (address → payment → confirm).	High
FR10	Payment Processing	The system shall integrate with a payment gateway to process credit card and online payments.	High
FR15	Admin – Product Management	Admin shall be able to add, edit, deactivate, and delete product listings.	High

FR16	Admin – User Management	Admin shall be able to view, suspend, or delete user accounts.	High
FR17	Admin – Order Management	Admin shall be able to view all orders, update statuses, and process refunds.	High

3.2 Non-Functional Requirements

The following table defines the quality attributes and constraints of the system:

ID	Category	Requirement	Metric
NFR01	Performance	System response time for product search and page load.	< 2 seconds (95th percentile)
NFR02	Performance	System must support concurrent users without degradation.	Up to 10,000 concurrent users
NFR03	Scalability	Platform architecture must scale horizontally as user load increases.	Auto-scaling on cloud
NFR04	Availability	System must maintain high uptime excluding planned maintenance.	99.5% uptime SLA
NFR05	Security	All data in transit must be encrypted.	HTTPS / TLS 1.2+
NFR06	Security	User passwords must be stored securely.	Bcrypt hashing (cost ≥ 12)
NFR07	Security	Payment data must not be stored on the platform.	PCI-DSS compliance
NFR08	Usability	Platform must be responsive and accessible on mobile and desktop.	Mobile-first design
NFR09	Usability	First-time users should complete checkout without training.	≤ 5 minutes for checkout
NFR10	Maintainability	Codebase must follow MVC architecture with documented APIs.	Modular architecture
NFR11	Reliability	Failed payments must not result in duplicate charges.	Idempotent payment calls
NFR12	Portability	Application must run on major browsers.	Chrome, Firefox, Safari, Edge

4. Software Architecture

4.1 Architecture Style

Stashio adopts the Model-View-Controller (MVC) architectural pattern, separating the application into three interconnected components:

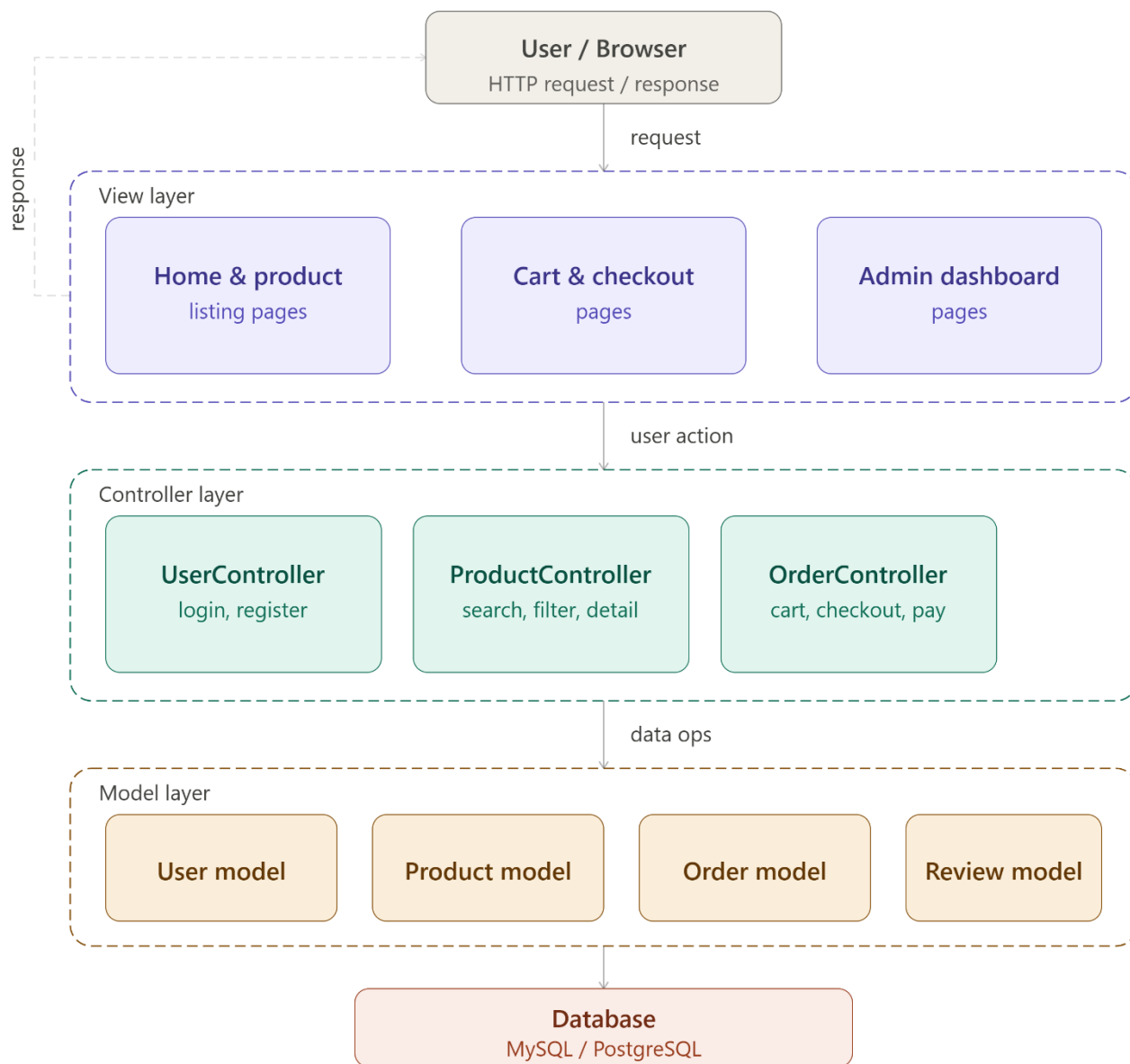
- Model – Manages data, business logic, and database interactions (Products, Users, Orders, Payments).
- View – Handles the user interface and presentation layer (HTML/CSS/React components).
- Controller – Processes user requests, coordinates between Model and View, and applies business rules.

The system uses a layered architecture with a REST API backend, allowing future extension to mobile apps or third-party integrations.

4.2 High-Level System Components

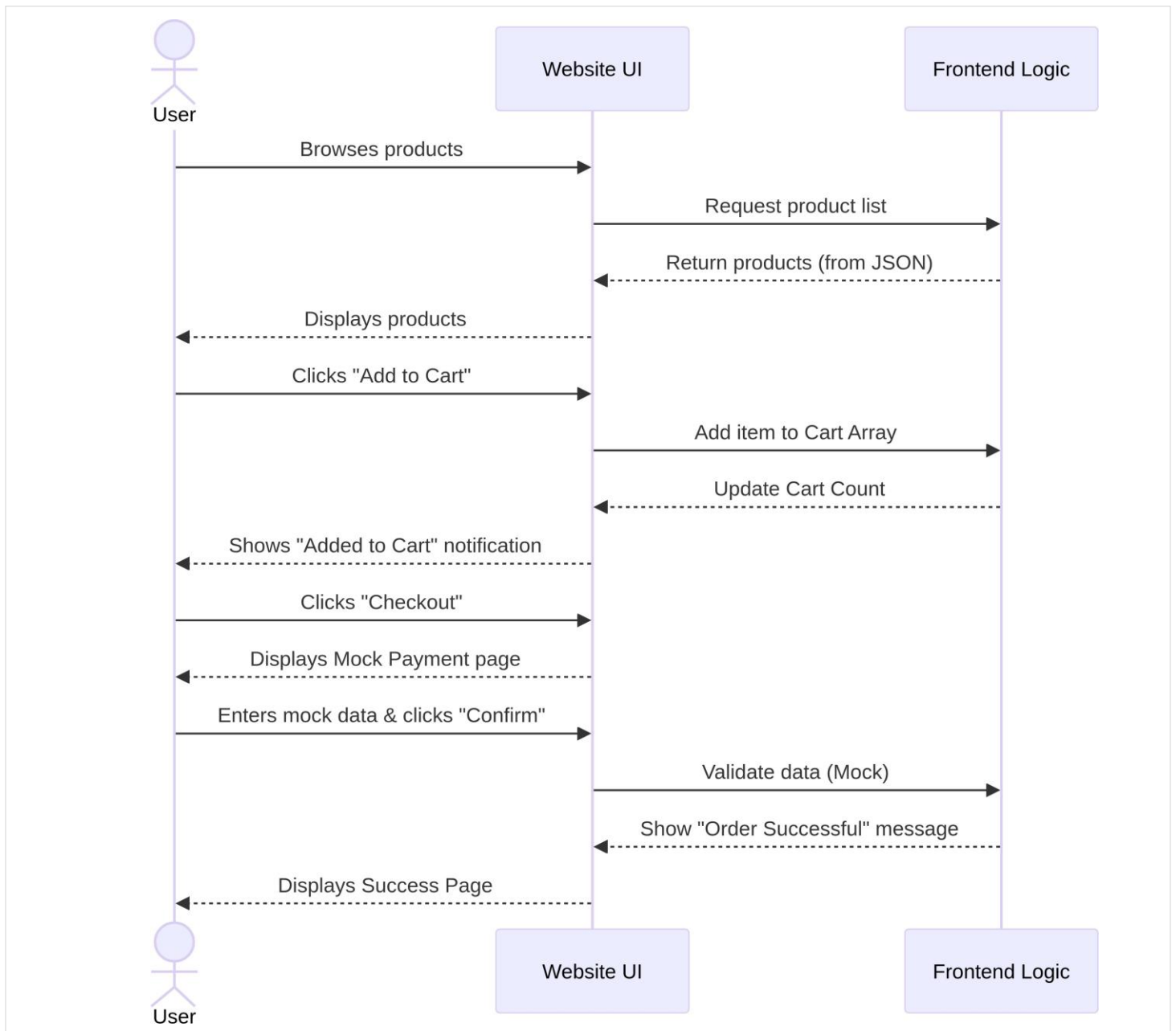
Component	Responsibility
Presentation Layer	React.js frontend – renders UI components (pages, forms, navigation)
API Gateway	Routes HTTP requests to the appropriate backend service/controller
Auth Service	Handles user registration, login, JWT token issuance and validation
Product Service	Manages product catalog, search, filtering, categories, and reviews
Order Service	Handles cart operations, order creation, status management, and history
Payment Service	Integrates with external Payment Gateway (e.g., Stripe/PayPal) for secure transactions
Admin Module	Dashboard for managing products, users, orders, and generating reports

4.3 Architecture Diagram

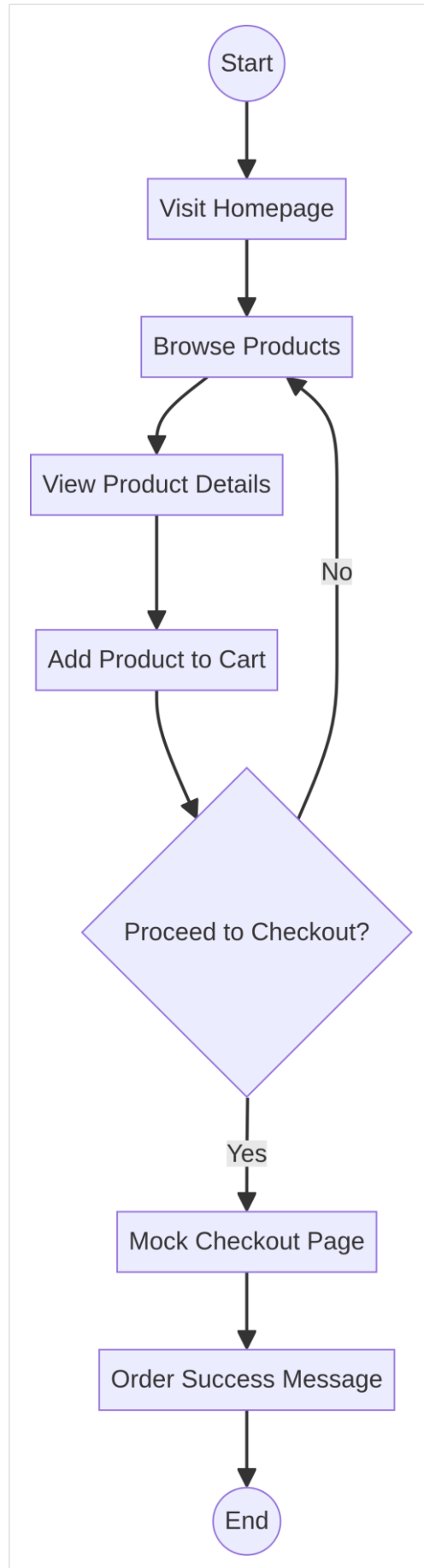


5. Data Flow & System Behavior

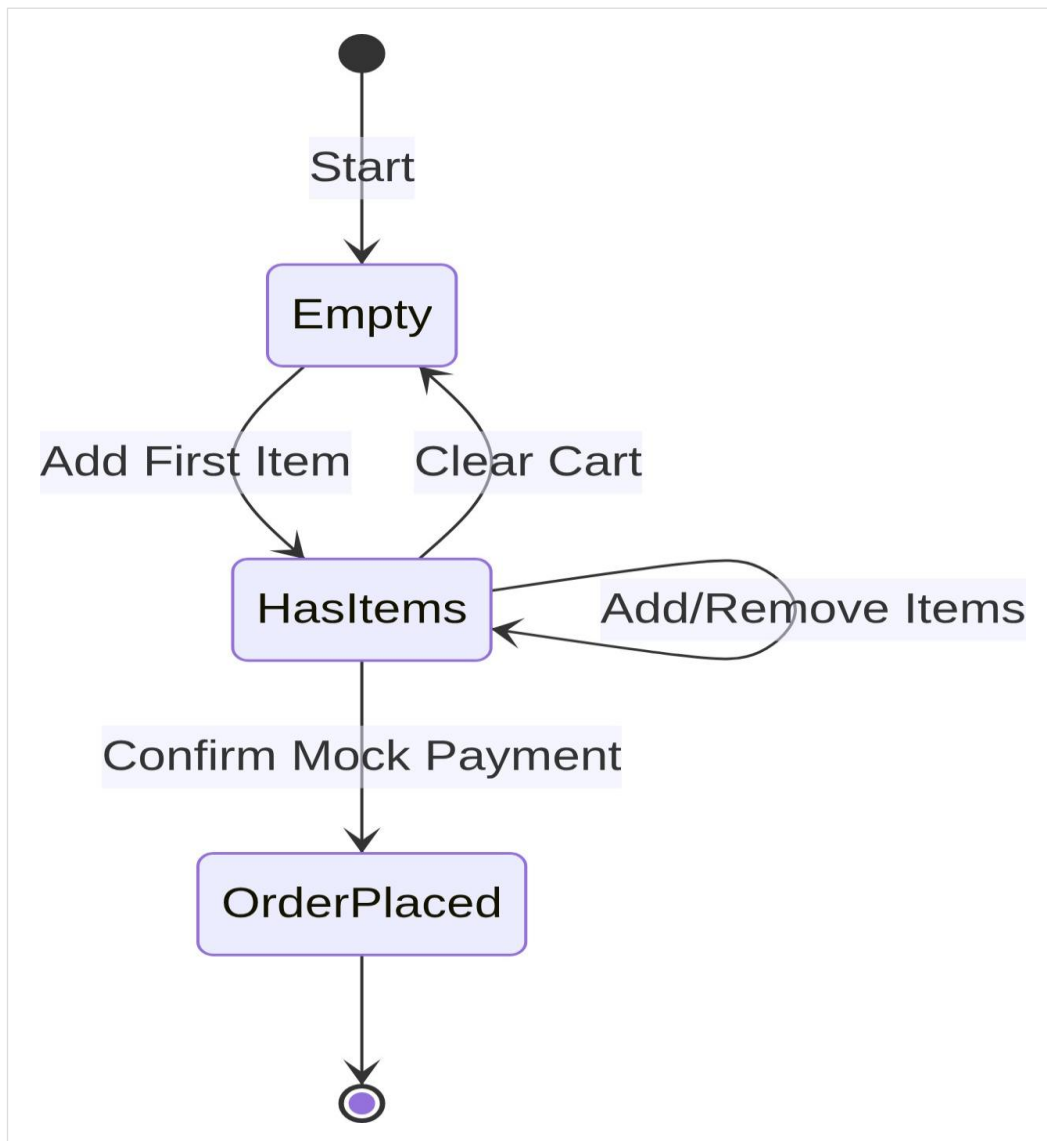
5.1 Sequence Diagram



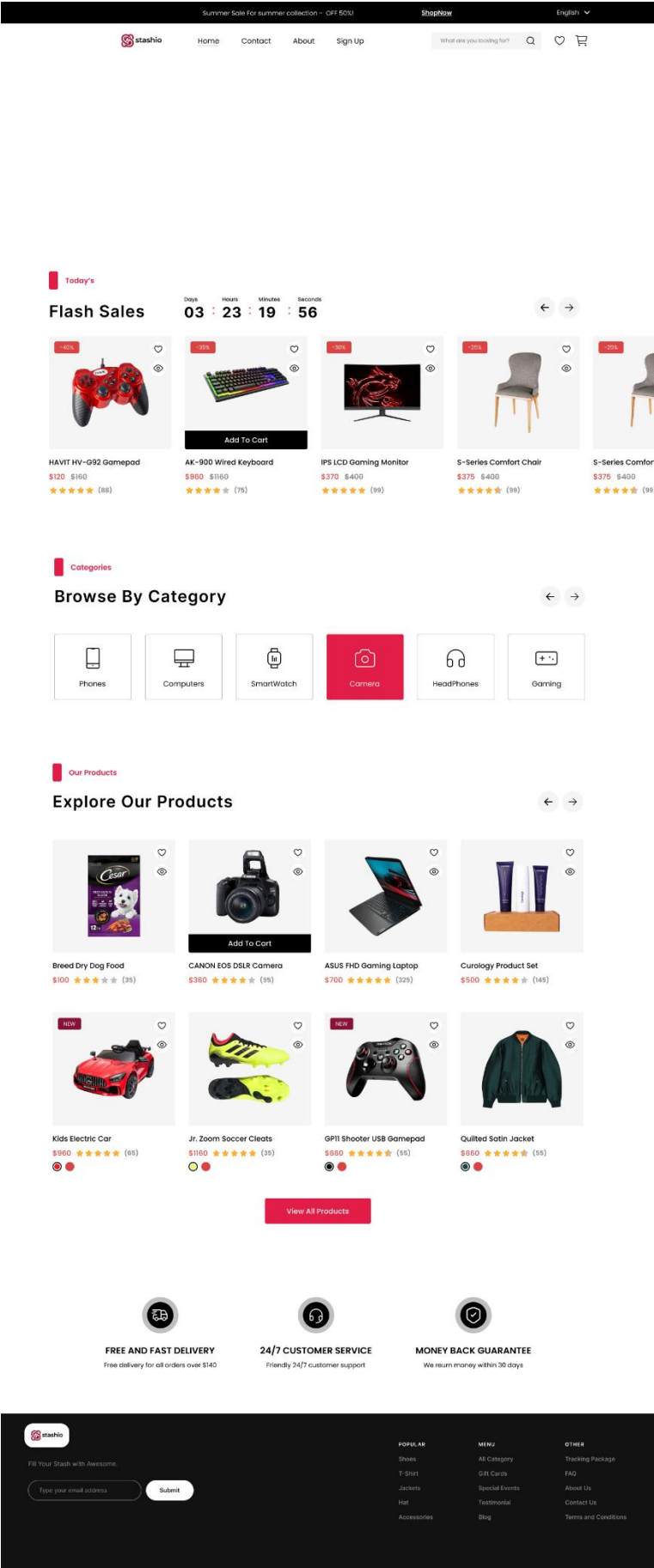
5.2 Activity Diagram



5.3 State Machine Diagram



6. UI/UX Design & Prototyping



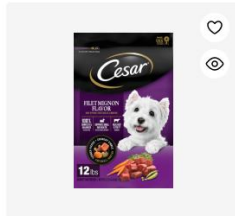
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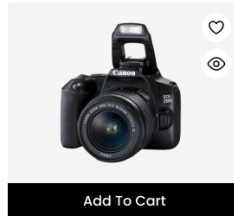
Our Products

Explore Our Products



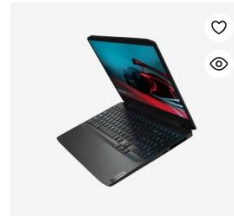
Breed Dry Dog Food

\$100 ★★★★★ (35)



CANON EOS DSLR Camera

\$360 ★★★★★ (95)



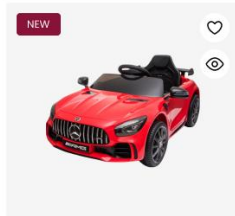
ASUS FHD Gaming Laptop

\$700 ★★★★★ (325)



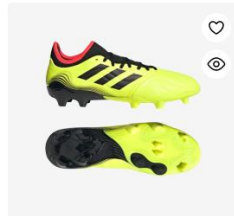
Curology Product Set

\$500 ★★★★★ (145)



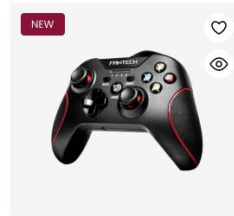
Kids Electric Car

\$960 ★★★★★ (65)



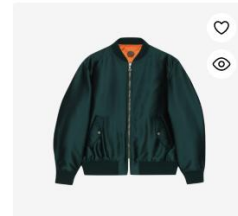
Jr. Zoom Soccer Cleats

\$1160 ★★★★★ (35)



GP11 Shooter USB Gamepad

\$660 ★★★★★ (55)



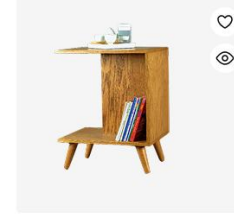
Quilted Satin Jacket

\$660 ★★★★★ (55)



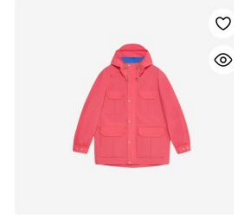
IPS LCD Gaming Monitor

\$1160 ★★★★★ (65)



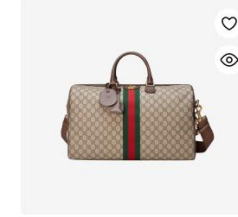
Small BookSelf

\$360 ★★★★★ (65)



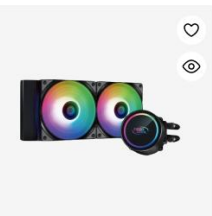
The north coat

\$260 ★★★★★ (65)



Gucci duffle bag

\$960 ★★★★★ (65)



RGB liquid CPU Cooler

\$160 ★★★★★ (65)



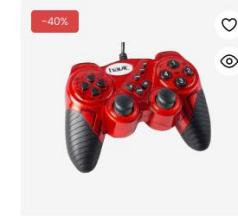
AK-900 Wired Keyboard

\$200 ★★★★★ (65)



S-Series Comfort Chair

\$375 ★★★★★ (99)



HAVIT HV-G92 Gamepad

\$120 ★★★★★ (88)



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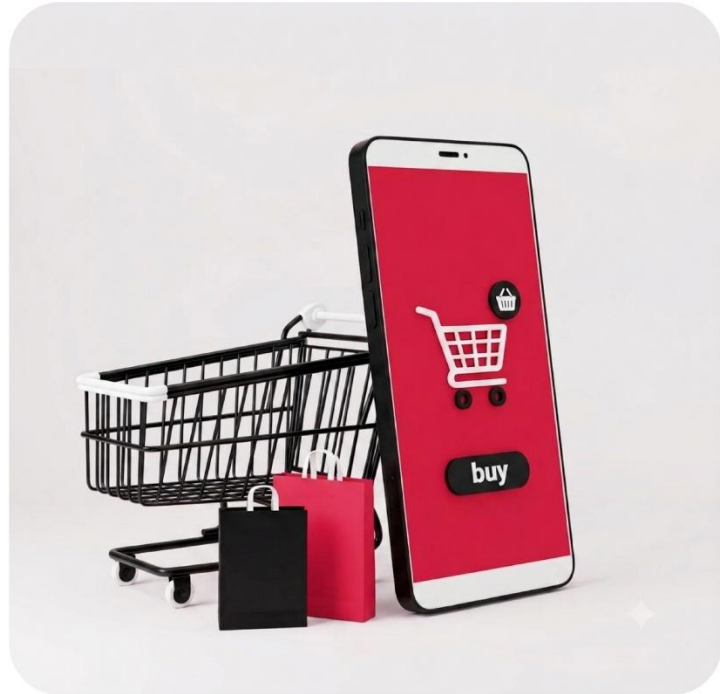
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Havic HV G-92 Gamepad

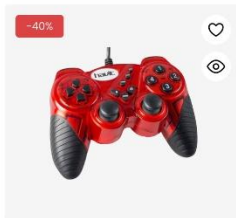
★★★★★ (150 Reviews) | In Stock

\$192.00

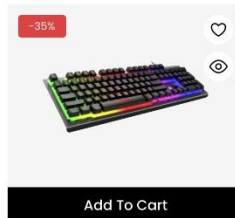
PlayStation 5 Controller Skin High quality vinyl with air channel adhesive for easy bubble free install & mess free removal Pressure sensitive.

Colours: Blue RedSize: XS S M L XL− 2 + Buy Now Heart

Related Item

**HAVIT HV-G92 Gamepad**~~\$120~~ \$160

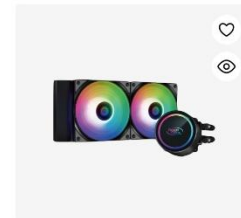
★★★★★ (88)

**AK-900 Wired Keyboard**~~\$960~~ \$1160

★★★★★ (75)

**IPS LCD Gaming Monitor**~~\$370~~ \$400

★★★★★ (99)

**RGB liquid CPU Cooler**~~\$160~~ \$170

★★★★★ (65)



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Product	Price	Quantity	Subtotal
 LCD Monitor	\$650	01	\$650
 H1 Gamepad	\$550	02	\$1100

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Cart Total

Subtotal: \$1750

Shipping: Free

Total: \$1750

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 Complete Your Order

Review your items and complete your purchase

 **Shipping Address**

Where should we deliver your order?

 **Saved Addresses**

Select a saved address or enter a new one below

 **Sadat City**
Sadat City
01XXXXXXX Sadat City

 **Use a different address**
Enter a new shipping address manually

 **Delivery Information**

Please ensure your address is accurate for smooth delivery

City *

e.g. Cairo, Alexandria, Giza

Street Address *

Street name, building number, floor, apartment...

Phone Number *

01xxxxxxxx Egyptian numbers only

 **Order Summary**

4 items

	Woman Shawl 2 x 149 EGP	298
	Woman Standart Fit Knitted Cardigan 1 x 499 EGP	499
	Woman Shawl 1 x 298 EGP	298
Subtotal		1,994 EGP
 Shipping		FREE
Total		1,994 EGP
 Place Order		

 Secure

 Fast Delivery

 Easy Returns

 **Payment Method**

Choose how you'd like to pay



Cash on Delivery
Pay when your order arrives at your doorstep





Pay Online
Secure payment with Credit/Debit Card via Stripe



☐

 **Secure & Encrypted**

Your payment info is protected with 256-bit SSL encryption



Create an account

Enter your details below

Name

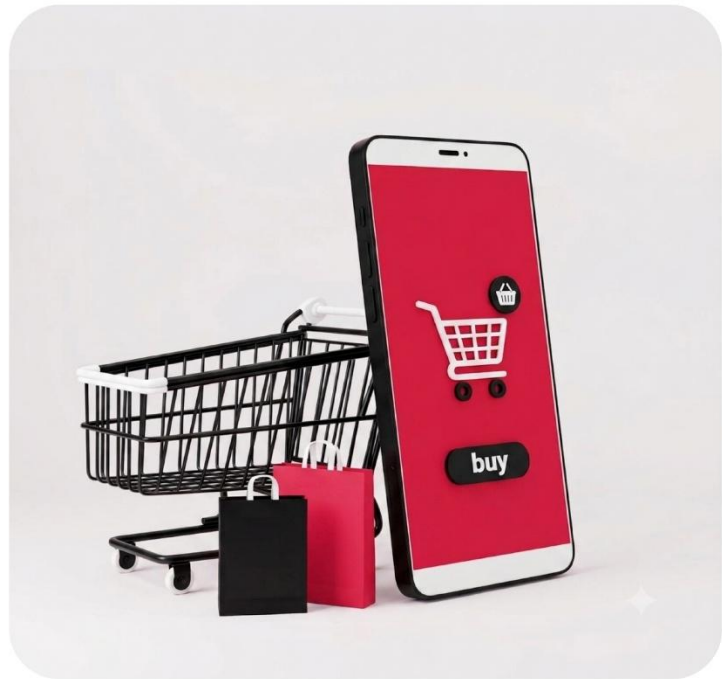
Email

Phone number

Password

Create Account

Already have account? [Log in](#)



Figma Link : <https://www.figma.com/design/cvmReuQxg1bJQSRtOcj4lt/Stashio-UI?node-id=49-3257&t=gs5crwCsNx59uqiP-1>

5. System Deployment & Integration

This section describes the technologies used to develop the E-Commerce Website and explains how the system components are deployed and integrated. The system follows a modern cloud-based architecture utilizing React for the frontend and Firebase services for backend functionality, authentication, database management, storage, and hosting.

5.1 Technology Stack

The E-Commerce Website is implemented using a modern web technology stack designed to ensure scalability, maintainability, responsiveness, and security.

Layer	Technology	Purpose
Frontend	React.js	Development of dynamic and interactive user interfaces
Styling Framework	Tailwind CSS	Responsive and utility-first styling
Routing	React Router	Global state management (cart, auth)
Authentication	Firebase	User registration, login, and session management
Database	Firebase	Cloud-based NoSQL database for application data
Storage	Firebase	Storage of product images and media assets
Hosting	Firebase	Deployment and hosting of the web application
Version Control	Git & Github	Source code management and team collaboration

Rationale for Key Choices

- Frontend

The frontend is developed using React.js as a Single Page Application (SPA). React provides reusable components and efficient rendering mechanisms that improve maintainability and performance. Tailwind CSS is utilized to create responsive and visually consistent user interfaces across different screen sizes.

- Backend Services

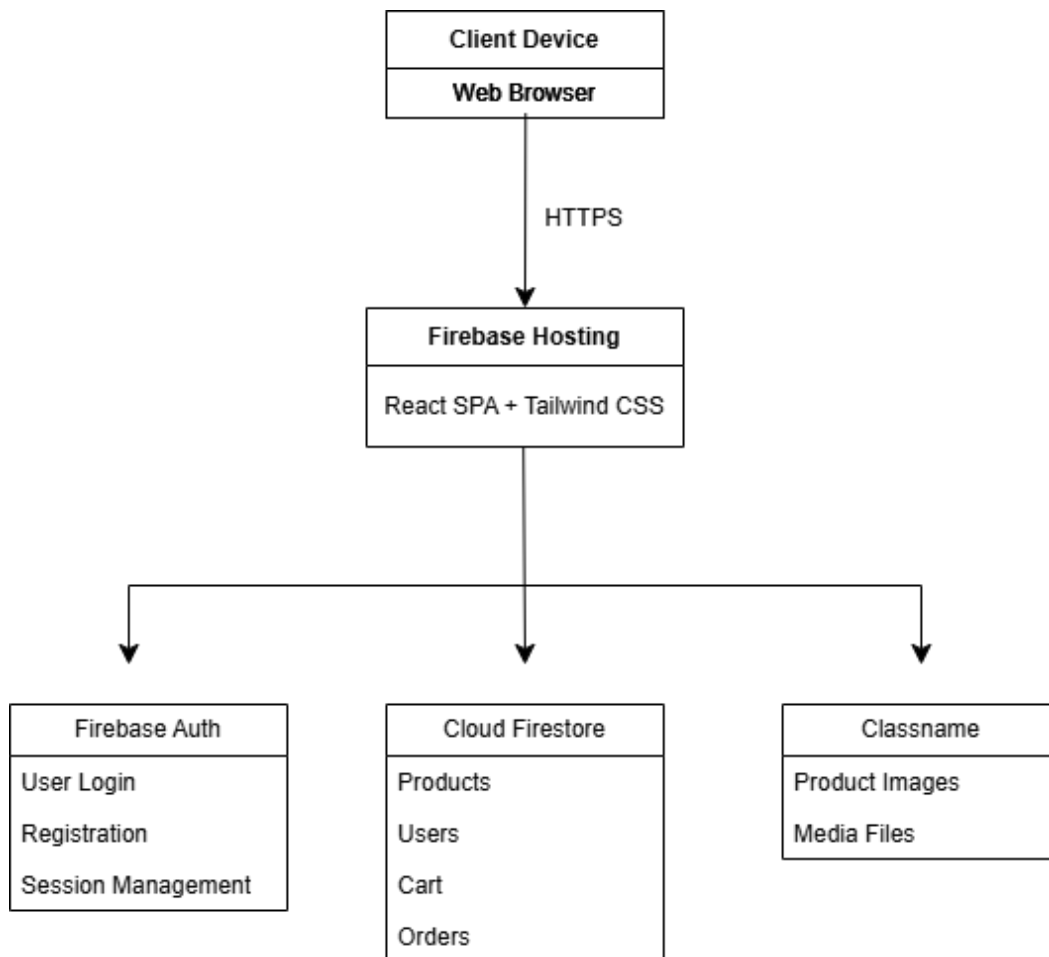
Firebase Authentication manages user accounts and secure login functionality. Firebase services eliminate the need for a traditional backend server while providing secure and scalable cloud-based services.

- Database

Cloud Firestore serves as the primary database for storing user information, products, shopping carts, and orders. The database supports real-time synchronization and automatic scalability.

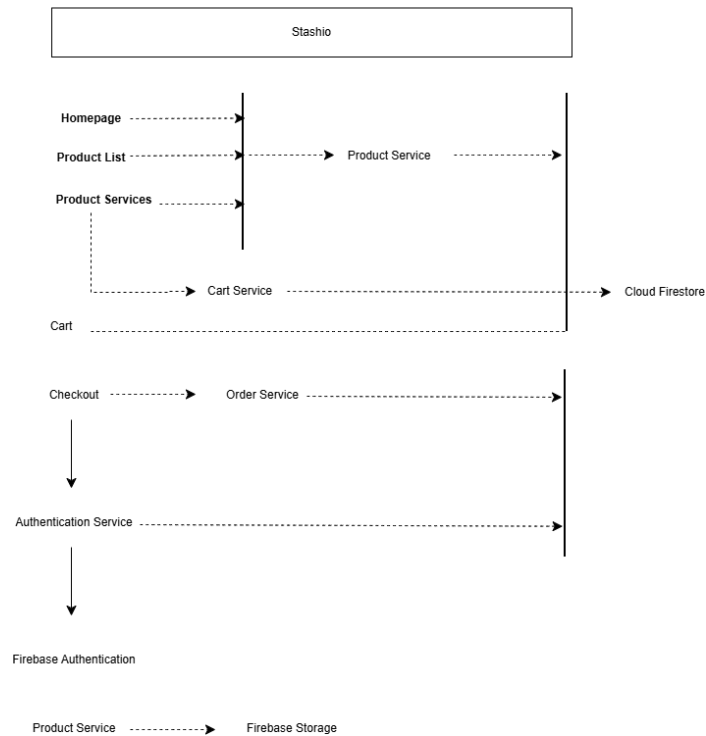
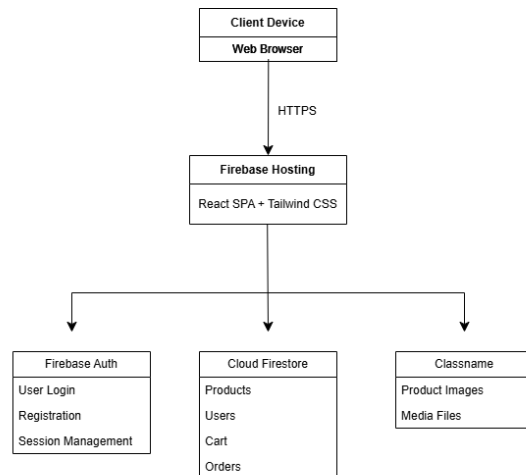
5.2 Deployment Diagram

The deployment architecture follows a cloud-based model. Users access the system through a web browser, which loads the React Single Page Application hosted on Firebase Hosting. Authentication services are managed through Firebase Authentication, application data is stored in Cloud Firestore, and product images are stored in Firebase Storage. All communication occurs over secure HTTPS connections.



5.3 Component Diagram

The component diagram illustrates the modular structure of the e-commerce application. User interface components interact with service components responsible for authentication, product management, cart operations, and order processing. These services communicate with Firebase Authentication, Cloud Firestore, and Firebase Storage to provide the application's backend functionality.



Component	Responsibility
Homepage Component	Displays hero banner and featured products
Product List Component	Displays searchable and filterable product catalog
Product Details Component	Displays detailed product information and allows adding items to cart
Cart Component	Manages shopping cart operations
Checkout Component	Handles checkout workflow and order placement
Authentication Service	Handles login, logout, and user validation
Product Service	Retrieves and manages product data
Cart Service	Manages cart items and quantity updates
Order Service	Processes customer orders
Firebase Authentication	User authentication and session management
Cloud Firestore	Stores products, users, carts, and orders
Firebase Storage	Stores product images

6. Class Diagram

The UML Class Diagram illustrates the static structure of the E-Commerce Website by representing the main classes, their attributes, methods, and relationships. The **User** class represents registered customers who can manage a shopping cart and place orders. Each user owns a single **Cart**, which contains one or more **CartItem** objects. Each cart item is associated with a specific **Product**, storing the selected quantity and subtotal information.

The **Order** class represents completed purchase transactions created by users during checkout. Every order includes exactly one **Payment** record and one **ShippingInfo** record, ensuring that payment details and delivery information are stored separately for better organization and maintainability.

The diagram uses associations and composition relationships to model dependencies between classes. Multiplicity constraints indicate the number of instances that can participate in each relationship, providing a clear representation of the system's data model and object interactions. This design supports the core functionalities of the e-commerce platform, including product browsing, cart management, order processing, payment handling, and shipping management.

